

When does an AI lead become lasting power?

Advanced AI, the US–China relationship, and possible worlds through 2040.

Conversations in Washington and my experience at the World AI Conference in Shanghai kept bringing me back to the same question: if a country gets better AI first, what does it get to do that others cannot?

I want to take the compounding case seriously. If AI improves the research that produces its successors—and helps redesign the machinery around it—a lead could become much more than an advantage in chat software. The difficult question is whether control over that process stays exclusive long enough to become durable power.

6 September 2026 · private-review prototype · draft personal copy for Jinhua’s edit.
GitHub materials are public.

Three clocks, no fixed winner.

Advance

Does AI improve the process of making the next AI?

Spread

Who gets usable capability before exclusivity pays?

Convert

Which firms and institutions turn it into consequential work?

These processes can change one another. A technological lead, control over deployment and broad social benefit can end up in different hands.

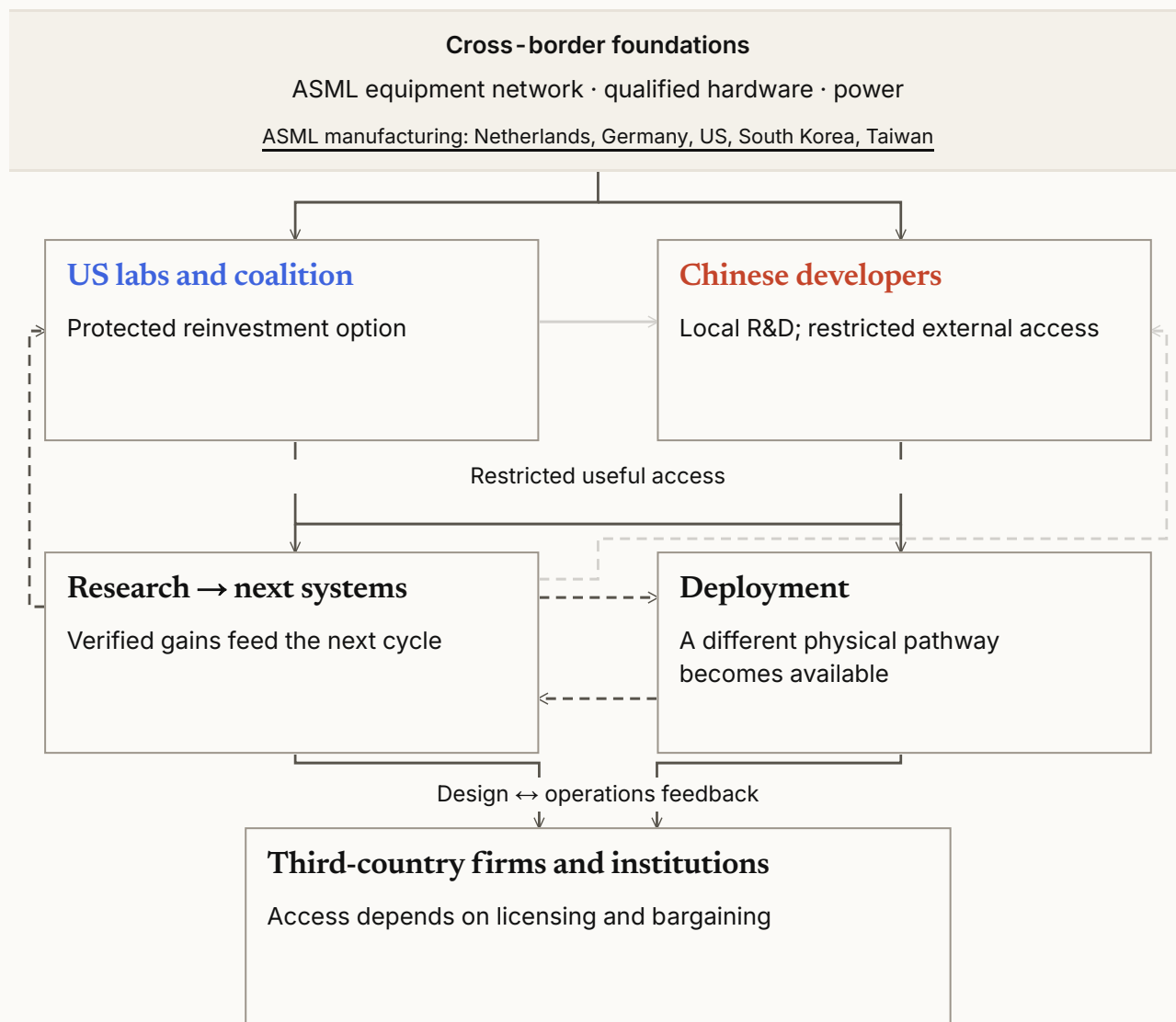
Start with the strongest compounding case.

Imagine a US-based frontier coalition gets a lead in research-relevant AI. It protects that access, repeatedly uses AI to improve its successors, and brings validated process changes into physical deployment. Give it those conditions before asking what could undo the advantage. The initial national lead is a stipulation, not a ranking.

The canvas stays beside the unfolding argument. Change the conditions or choose a proposed future, then compare it with the pinned starting case.

Who can turn a lead into power?

Conditional network · compare the outlined changes with the pinned starting case.



Owners + selected state users

Output can grow while gains remain concentrated

Scenario connections, not measured effects. Both actor groups research and deploy. Dashed lines: changeable feedback; faded: inactive or unassessed. Outlines mark changes. Position, color and line width are not power scores.

Now: A protected research lead can compound. Output can grow while gains remain concentrated.

Inspect rules, evidence and rival accounts

The lead compounds

Give the frontier case its strongest conditions: repeated research advances, protected access and validated changes to the production path.

Draft scenario, not an approved forecast. Dates below are stipulated windows, not an estimated chronology.

2026 → opening premise

A lead worth protecting

Stipulate a US-based frontier coalition with an initial advantage in the capabilities relevant to this story. Chinese developers pursue their own systems; firms in other countries can buy, adapt or bargain for access. This is a chosen starting premise, not a measured ranking of today's models.

Before 2030 · stipulated window

Research changes the next research cycle

Assume the systems help researchers generate, test and retain improvements across successive model-development cycles. This goes beyond the bounded AlphaEvolve report: repetition, secure reinvestment and reliable evaluation are additional premises. Assume these gains free scarce researcher time or computing work for the next cycle, and that the owners choose to reinvest rather than consume the savings. Better systems could help design algorithms, experiments and equipment; those improvements then support further research. This is the proposed compounding mechanism, not a rate inferred from one reported gain.

The evidence beneath the research premise

Change the route, not only the pace

Assume AI-assisted engineering produces validated changes to industrial integration, laboratory operations and equipment design. Dedicated capital, qualified hardware and operating staff are still required. The changed network includes deployment feedback into design: the scenario can create new ways around today's constraints rather than making every future wait at the same queue.

A physical-control example, with its limits

A protected research lead can compound

Restricted access plus repeated research feedback preserves an option for the initial coalition to concentrate scientific and strategic capacity. It could set access terms for important research tools, choose which discoveries to develop, and give selected public agencies earlier access to useful systems. Those are potential bargaining advantages, not a measured military edge. Allied suppliers retain bargaining power; dependence has not disappeared. The first beneficiaries are the owners and selected institutions controlling deployment. Public welfare does not follow automatically from geopolitical leverage.

What this bargain gives up

Assume a finite discretionary investment and implementation agenda. Moving attention to one priority reduces what can be attempted elsewhere. Money, qualified equipment and people are not interchangeable.

First call: Frontier experiments

Next call: Selected deployments

Deferred at the margin: Access and assurance

No additional national resource stock is created by a selector. A new deployment route requires dedicated capacity; the prototype does not calculate whether a real country can afford it.

Before / after

Starting case pinned; change a condition to compare.

Conditional implications under the same rules

Question	Before	Now
Strategic control	A protected research lead can compound	A protected research lead can compound
Physical output	A different physical pathway becomes available	A different physical pathway becomes available
Household benefit	Output can grow while gains remain concentrated	Output can grow while gains remain concentrated

What would make you change your mind?

These tests follow the selected assumptions. Evidence supporting a premise can weaken a rival branch. Under interruption, research and diffusion tests apply only if continuity returns.

Does the research loop repeat? Support: Successive independently evaluated improvements improve the end-to-end process of developing useful successors. Weaken: Gains stay confined to easily scored tasks, fail to reproduce, or consume the savings in evaluation and integration.

Does useful capability arrive before exclusivity pays? Support: The decisive research/security capability remains unavailable, or serving and qualified hardware prevent effective use. Weaken: Affordable, reliable substitutes reach relevant users before the leader can consolidate the selected advantage.

Does control stay concentrated? Support: Ownership, procurement and bargaining channel gains toward a limited group while broader household gains lag. Weaken: Reliable evidence shows widely shared gains through prices, wages, services or access.

The four rules behind this construction

Research feedback

Repeated advances activates the proposed research-to-next-system feedback. Bounded improvements leaves useful AI in research but does not close a self-reinforcing loop. Neither option supplies a rate.

Useful capability reaches rivals

Earlier diffusion gives Chinese developers and third-country adopters access to the relevant usable capability. It weakens the exclusive-access route to leverage, without transferring chips, trained staff, ownership or all frontier research capabilities for free.

AI changes the production path

Validated redesign adds a proposed route from AI-assisted engineering into repeatable physical deployment and feedback from operations. It stipulates that validation, equipment and people arrive together; existing-path mode retains separate qualification and integration work. No stage is removed silently.

Investment and political bargain

This bundled choice shifts a fixed discretionary agenda among frontier experiments, deployment and access/assurance. It changes ownership and procurement assumptions as well as priorities. It is not a measured national trait. A compact requires verification and cooperation; rupture withdraws benign outcome claims.

Full assumption and source trace

The default construction is readable without JavaScript. Controls need JavaScript. This is an executable qualitative branch explorer, not a calibrated simulation.

The starting evidence points in more than one direction.

AI improving the tools that build AI

Google reports using AlphaEvolve to improve a kernel used in Gemini training, reducing total training time by 1%.

Originating report · Google DeepMind · 2025-05-14

A channel for useful capability to spread

DeepSeek released R1/R1-Zero weights and six distilled models. Released weights make local use and adaptation possible.

Originating report · DeepSeek · 2025-01-22

China's programme reaches from models into production

The AI+ opinion calls for stronger foundational models, integration into production workflows, application testing bases and shared industry platforms.

Official programme intent · State Council; official National Data Administration reproduction · 2025-08-26

The US programme also includes diffusion

The AI Action Plan proposes open-weight support, organizational adoption, infrastructure and exports spanning hardware, models, software and services.

Official programme intent · White House · 2025-07-23

A laboratory does not own its entire supply network

ASML reports specialized single-source dependencies, including Carl Zeiss SMT optics, and manufacturing operations across the Netherlands, Germany, US, South Korea and Taiwan.

Originating report · ASML · 2026-02-25

Physical deployment can change, too

IEA identifies grid connections and equipment lead times as possible constraints and discusses efficiency, flexibility and siting as mitigations.

Source analysis · International Energy Agency · 2025-04-10

Neither national programme fits a simple “US invents, China deploys” story. Their intentions overlap; which actors deliver, at what cost and for whose benefit is an empirical question. [Read the exact limits, including the draft Chinese translation.](#)

Power for whom?

I care about who gets to decide what this power is for. A state could become harder to coerce while many of its citizens gain little. A society could become healthier or more capable without owning the leading laboratory. I want the comparison to keep both possibilities in view.

The race branch can concentrate productive and strategic capability. Earlier diffusion can spread access without spreading ownership. A public mission can turn assurance into a condition for useful deployment, but verification can fail and priorities have costs. Conflict or loss of effective control can break the entire benign trajectory. None of these possibilities is settled by a workplace productivity percentage.

Draft author interpretation. No scenario probability or publication assent recorded.

Supporting study: what AI changed in actual work · AI 2027, Plan A and AI as Normal Technology · Authorship

